

Pre-Board Examination- Class 10

MATHEMATICS (Standard)

Time Allowed: 3 hrs.

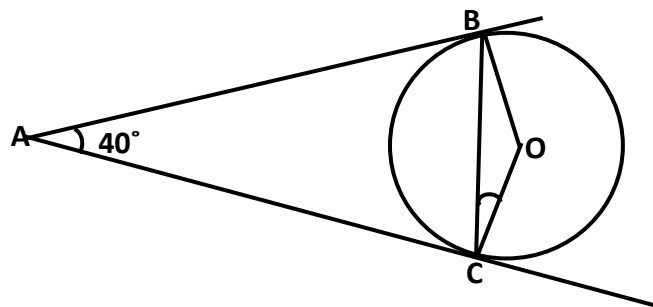
Maximum Marks: 80

General Instructions:

1. This Question Paper has 5 sections A to E.
2. All Questions are compulsory. However, internal choice is given in some questions.
3. Section A has 20 MCQs having 1 mark each.
4. Section B has 5 Questions having 2 marks each.
5. Section C has 6 Questions having 3 marks each.
6. Section D has 4 Questions having 5 marks each.
7. Section E has 3 Case based integrated units (having 4 mark each) with sub parts of values of 1, 1 and 2 marks each respectively.
8. Draw neat figures wherever required.
9. Use the value of $\pi = (22/7)$ if otherwise not specified.

Section A

1. Find the value of k for which the system of equations $x + (k + 1)y = 5$ and $(k + 1)x + 9y = 8k - 1$ has infinitely many solutions
 (a) 2 (b) 3 (c) 4 (d) 5
2. In the given figure, AB and AC are tangents to a circle with centre O such that $\angle BAC = 40^\circ$, then $\angle BOC$ is equal to



- (a) 40° (b) 50° (c) 140° (d) 150°
3. $\triangle ABC \sim \triangle PQR$. If AM and PN are altitude of $\triangle ABC$ and $\triangle PQR$ respectively and $AB^2 : PQ^2 = 25 : 36$, then $AM : PN = \text{-----}$
 (a) 2 : 5 (b) 5 : 6 (c) 25 : 36 (d) 2 : 3
 4. If the area of a sector of a circle is $\frac{2}{20}$ of the area of the circle, then the angle at the centre is equal to

- (a) 30° (b) 36° (c) 45° (d) 60°

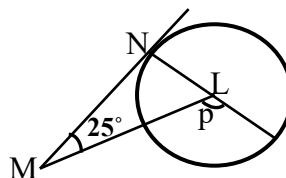
5. If the circumference of a circle increases from 2π to 4π , then its area becomes ----- of the original area
 (a) half (b) double (c) three times (d) four times
6. P (5,1), Q (1,4) and R (8,5) are the coordinates of the vertices of a triangle. Which of the following types of triangle will ΔPQR be
 (a) equilateral triangle (b) scalene right angled triangle
 (c) isosceles right angled triangle. (d) isosceles acute angled triangle.
7. If -2 and 3 are the zeroes of quadratic polynomial $x^2 + (p + 1)x + q$, then p and q are ---- - respectively.
 (a) $-7, 2$ (b) $-6, -2$ (c) $2, 6$ (d) $-2, -6$
8. A point (x, y) is at a distance of 7 units from the origin. How many such points lie in the third quadrant?
 (a) 0 (b) 1 (c) 2 (d) Infinitely many
9. Two right circular cones have their heights in the ratio $1 : 3$ and the radii in the ratio $3 : 1$. What is the ratio of their volumes ?
 (a) $1 : 3$ (b) $2 : 3$ (c) $3 : 1$ (d) $9 : 1$
10. For the following distribution, the sum of upper limits of the median and modal class is

Class	0-5	5-10	10-15	15-20	20-25
Frequency	11	28	24	18	19

- (a) 25 (b) 20 (c) 15 (d) 10
11. If $g - 1$, $g + 3$ and $3g - 1$ are in A.P, then find the value of g .
 (a) 3 (b) 4 (c) 5 (d) 7
12. If mode and mean of a data set are given as 8 and 9 respectively, then the median of the data set is
 (a) 8.57 (b) 8.67 (c) 8.97 (d) 9.24

13. In the given figure, MN is a tangent to the circle with centre L. Find value of $\angle p$.

- (a) 110°
 (b) 115°
 (c) 120°
 (d) 125°



14. Find the 11th term from the end of the given progression 10, 7, 4,, -62 .

- (a) -32 (b) -16 (c) -25 (d) 0

15. A single letter is selected at random from the word PROBABILITY. The probability that the selected letter is a vowel is -----.

- (a) $\frac{2}{11}$ (b) $\frac{3}{11}$ (c) $\frac{4}{11}$ (d) $\frac{11}{4}$

16. In triangles ABC and DEF, $\angle B = \angle E$, $\angle F = \angle C$, and $AB = 3DE$. Then the two triangles are

- (a) congruent but not similar (b) similar but not congruent
(c) neither congruent nor similar (d) congruent as well as similar

17. What is the HCF of the least prime number and the least composite number

- (a) 1 (b) 2 (c) 3 (d) 4

18. $\sqrt{n}$ is a natural number such that $n > 1$. Which of these can definitely be expressed as product of primes?

- (i) $\sqrt{n}$ (ii) n (iii) $\frac{\sqrt{n}}{2}$

- (a) only (i) (b) only (i) and (ii) (c) all (i), (ii) and (iii) (d) None of these

Question number 19 and 20 are Assertion and Reason based questions carrying 1 mark each. Two statements are given, one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to the questions from the codes (a), (b), (c) and (d).

19. Statement A (Assertion): HCF of 26 and 91 is 13 and the LCM is 182.

Statement R (Reason): As $\text{HCF}(a,b) \times (a \times b) = \text{LCM}(a,b)$

- (a) Both Assertion (A) and (R) are True and Reason (R) is the correct explanation of Assertion (A)
(b) Both Assertion (A) and Reason (R) are True but Reason (R) is not the correct explanation of Assertion (A)
(c) Assertion (A) is True but Reason (R) is False
(d) Assertion (A) is False but Reason (R) is True

20. Assertion (A): The mid-point of a line segment divides it in ratio $1 : 1$.

Reason (R): Three points are collinear if they lie on same straight line.

- (a) Both Assertion (A) and (R) are True and Reason (R) is the correct explanation of Assertion (A)
(b) Both Assertion (A) and Reason (R) are True but Reason (R) is not the correct explanation of Assertion (A)
(c) Assertion (A) is True but Reason (R) is False
(d) Assertion (A) is False but Reason (R) is True

Section B

21. Solve for x and y : $71x + 37y = 253$
 $37x + 71y = 287$

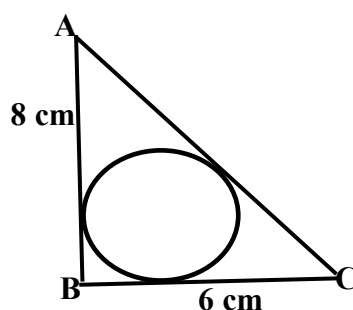
OR

Find value of x and y. $26x + 28y - 269 = 0$; $28x + 26y - 271 = 0$.

22. Volumes of two spheres are in the ratio 27 : 64, then find the ratio of their surface areas.
23. Prove that the tangents drawn from an external point to a circle are equal in length.
- OR**
- Prove that the parallelogram circumscribing a circle is rhombus.
24. A cow is tied to a peg at one corner of a rectangular shaped grass field of 20 m long and 16 m wide with a 14 m long rope. Find the area of that part the cow cannot graze.
25. Find a quadratic polynomial whose zeroes are $(3 + \sqrt{3})$ and $(3 - \sqrt{3})$.

Section C

26. Show that $\frac{(\sin A - 2\sin^3 A)}{(2\cos^3 A - \cos A)} = \tan A$.
27. Given that $\sqrt{3}$ is irrational. Prove that $1 - 2\sqrt{3}$ is irrational.
28. A box contain 90 paper discs which are numbered from 1 to 90. There is only one number on a disc. If one disc is drawn at random from the box find the probability that it bears
- (i). a two-digit number.
 - (ii). a perfect square number.
 - (ii). a number divisible by 5.
29. $\triangle ABC$ is right-angled at B with a circle inscribed in it. The lengths of two sides containing the right angle are 6 cm and 8 cm. Find the radius of the circle.



30. A lending library has a fixed charge for first three days and an additional charge for each day thereafter. Priya paid ₹ 50 for a book kept for eight days. Her friend Shilpa paid ₹ 85 for a book kept for thirteen days. Find the fixed charge and the charge for each extra day.

OR

The numerator of a fraction is 4 less than the denominator. If the numerator is decreased by two and the denominator is increased by one, then the denominator is 8 times the numerator. Find the fraction.

31. A sphere of diameter 12 cm is dropped in a right circular cylindrical vessel filled with water. If the sphere is completely submerged in the water, the water level in the cylindrical vessel rises by $3\frac{5}{9}$ cm. Find the diameter of the cylindrical vessel.

OR

A wooden toy is in the shape of a cone standing on a hemisphere with both their radii being equal to 3 cm and the height of the cone is equal to 5 cm. Find the volume of the wood used.

Section D

32. The monthly consumption of electric power of 68 consumers of a locality under Harippad Electricity office is shown in the following table. Find the mode and median of the data.

Monthly consumption (in KWh)	65 - 85	85 - 105	105 - 125	125 - 145	145 - 165	165- 185	185- 205
Number of consumers	4	5	13	20	14	8	4

OR

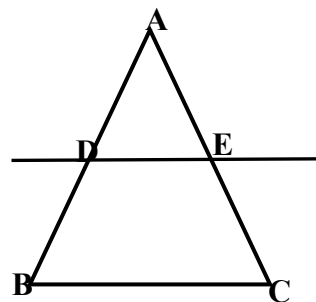
If the mean of the following frequency distribution is 91, then find the value of f_1 and f_2 .

Class	0-30	30-60	60-90	90-120	120-150	150-180	Total
Frequency	12	21	f_1	52	f_2	11	150

33. Prove that, if a line segment is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, the other two sides are divided in the same ratio. Using the above result, prove the following.

In the given figure, $DE \parallel BC$ and $BD = CE$.

Prove that ΔABC is an isosceles triangle.



34. In a construction site where multistorey buildings are constructing, structures of 2 towers (named as Kaveri and Ganga) are completed. They are on same horizontal line. The angle of elevation of the top of 24 m high Kaveri tower from the foot of Ganga tower is 60° . The angle of elevation of top of Ganga tower from the foot of Kaveri tower is 30° . What is the distance between the two towers? Find the height of Ganga tower.

OR

A spherical balloon of radius r subtends an angle of 60° to its top for a viewer standing on ground. If the angle of elevation of its centre is 45° from the same point for the viewer, then find the height of the centre of the balloon from ground in terms of its radius.

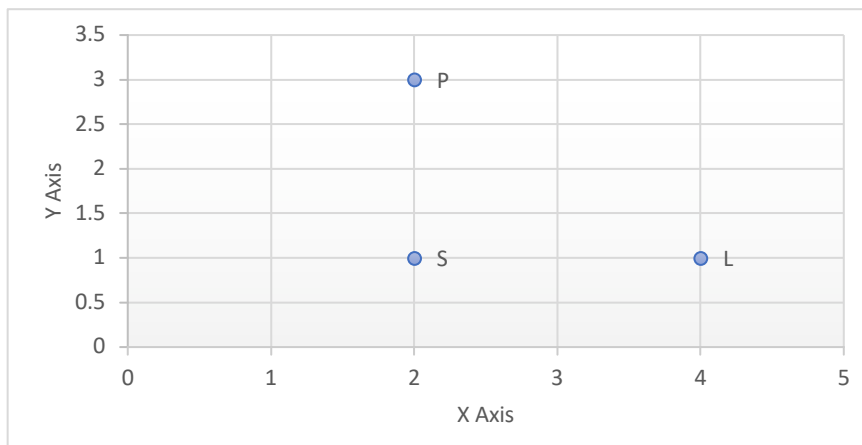
35. The ratio of 11th term to the 18th term of an Arithmetic Progression (A.P) is 2 : 3. Find the ratio of sum of its first five terms to sum of its first 21 terms.

Section E

36. The panchayat office (P) and School (s) are on the same street in a village. School and Library (L) building are also on a same street, but not the former street. The co-ordinates of the three buildings are marked on a graph paper, by taking 1 unit = 500 m, as shown below.

On the basis of given graph answer the following questions:

- (i) What is the distance between Panchayat and school building?
- (ii) How far is Library from the school?
- (iii) What type of triangle is formed by joining positions of Panchayat, School and Library?



37. India gate (formerly known as All India war memorial) is located near Karthavya path (formerly Rajpath) at New Delhi. It stands as a memorial to 74187 soldiers of Indian Army, who gave their life in the first world war. This 42 m (Approx. 138 feet) tall structure was designed by Sir Edwin Lutyens in the style of Roman triumphal arches. A group of students visited India gate as a part of their study tour.



- (i). What is the angle of elevation if they are standing at a distance of 42 m away from the India Gate?
- (ii). They want to see the monument at an angle 60° , then what will be the distance at which they should stand?
- (iii). If the altitude of the sun is 60° , then find the height of vertical tower that cast its shadow of 20 m long.

OR

The ratio of length of a pillar to its shadow is $1 : \sqrt{3}$. Find the angle of elevation of sun.

38. A carpenter wants to make a 3 m long ladder having rungs (steps) 25 cm apart. The rungs decrease uniformly in length from 45 cm at bottom to 25 cm at the top. The top and bottom rungs are at a distance of 2.5 m. Answer the following questions. Assume there is no wastage of wood.

- (i). Find the total number of rungs on the ladder.
- (ii). What is the length of the middle rung?
- (iii). Find the length required for making the ladder.

OR

Find the length required for the rungs.

